

PRAMAAN प्रमाण

System Architecture

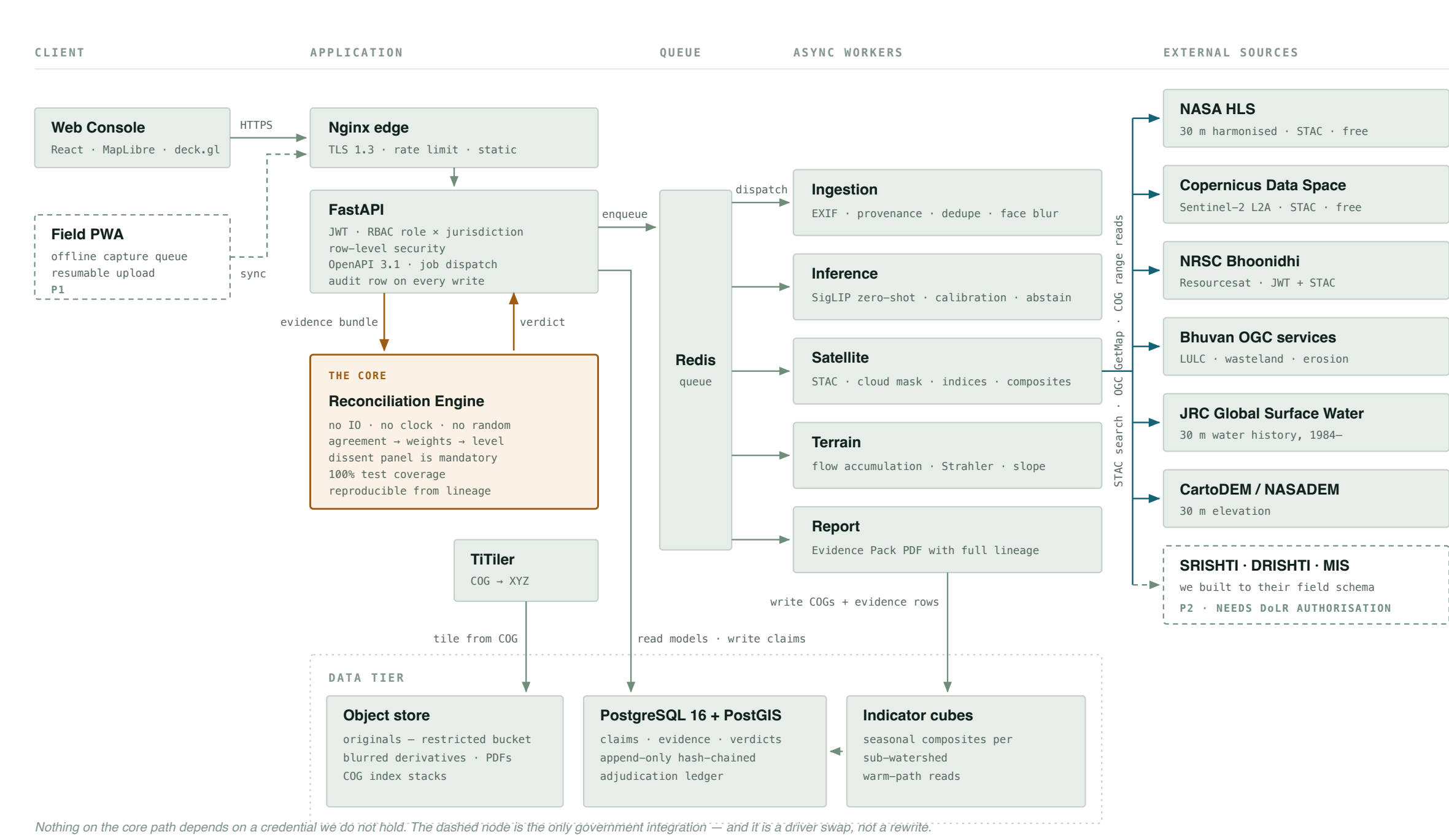
An evidence-reconciliation layer over the Department of Land Resources’ existing SRISHTI / DRISHTI geotag pipeline — turning each geo-coded watershed photograph into a claim the satellite record can test.

Client Ministry of Rural Development · Department of Land Resources

Scope Runtime architecture and the reconciliation mechanism

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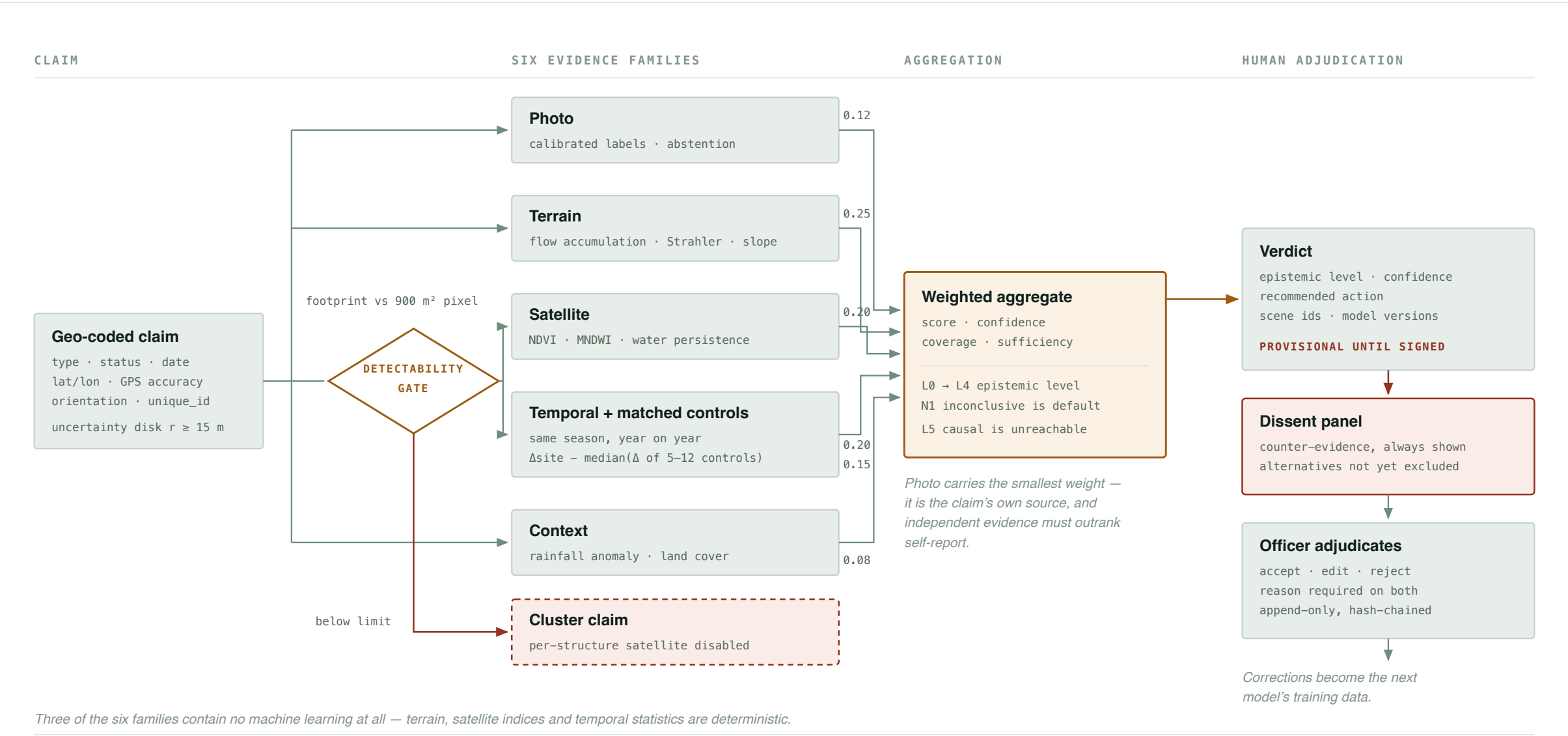
FIGURE 1 Runtime architecture, and what actually moves between tiers



Read it as three separable concerns. The application tier owns identity and jurisdiction and nothing else. The workers own everything slow — scene search, cloud masking, index compositing, hydrological derivation, inference, PDF generation — and every one of them is a retryable job with an id you can inspect. The amber box owns judgement, and it is

drawn with exactly two connections for a reason: an evidence bundle in, a verdict out, and no third arrow to a database, a clock or a network. **That constraint is the whole reason a machine-generated verdict can survive an audit**, because the same bundle always yields the same verdict and the stored lineage record is enough to recompute it years later. Every source drawn in teal is free, open and programmatically reachable today; the dashed node is the one thing we cannot reach without departmental authorisation, and the system is fully functional without it.

FIGURE 2 Inside the engine — how a photograph becomes an adjudicable verdict



The gate is the mechanism worth staring at. Before any satellite evidence is computed, the claimed structure's expected footprint is measured against the 900 m² area of a single 30 m pixel. A 25 × 25 m farm pond fails, and the amber path collapses: per-structure satellite evidence is switched off and the claim is escalated to a cluster assessment — visibly, in the interface and in the generated PDF. **That one deterministic branch is what stops the system from reporting a confident absence for something it was never able to see**, and it is precisely why a negative verdict from the families that survive can be trusted. Note also what the fan-in encodes: the photograph is the claim's own source, so it is weighted lowest of the six — the five independent families carry 0.88 against its 0.12. Terrain, satellite and temporal evidence are each produced without reference to the photograph, which is what makes their agreement informative — and what lets the system keep working at a lower ceiling when any single family is unavailable.

READING THE DRAWINGS



Amber — judgement

The pure reconciliation engine, the weighted aggregation, and the detectability gate that governs what the engine is even allowed to consider.



Teal — external data in motion

STAC queries, OGC requests and COG range reads against sources that are open and free today.



Clay — the negative path

A failed gate, the mandatory dissent panel, and every conclusion the system declines to reach.



Dashed — not built for the hackathon

The offline field client, and the government integration that requires DoLR authorisation.

Four decisions these drawings encode

The core has no IO

The engine is a pure function called by the API, drawn deliberately with only two arrows: a bundle in, a verdict out. No database handle, no clock, no network call. **The same bundle always produces the same verdict** — the only reason a machine-generated conclusion can be defended under audit years after it was issued.

The system is allowed to not know

Inconclusive is the default outcome, not a failure state. A cloud-blocked season, a missing baseline, or a structure below the sensor’s detection limit all resolve to **“requires physical verification”** — never to an accusation. That phrase is the strongest language available anywhere in the interface, the API or the generated report.

One arrow into the government workflow

PRAMAAN inserts itself between **geo-tag** and **moderate** in DoLR’s existing chain, and touches nothing else. It adopts the government’s own `unique_id`, its role vocabulary — SLNA, WCDC, PIA, WDT — and its work-status ladder. SRISHTI, DRISHTI and Bhuvan keep doing what they already do well.

Independence is what the weights buy

Terrain, satellite and temporal evidence are each generated without reference to the photograph. **That is what makes their agreement mean something**, and what lets the system degrade gracefully: lose a family and it says so, lowers its own ceiling, and carries on.